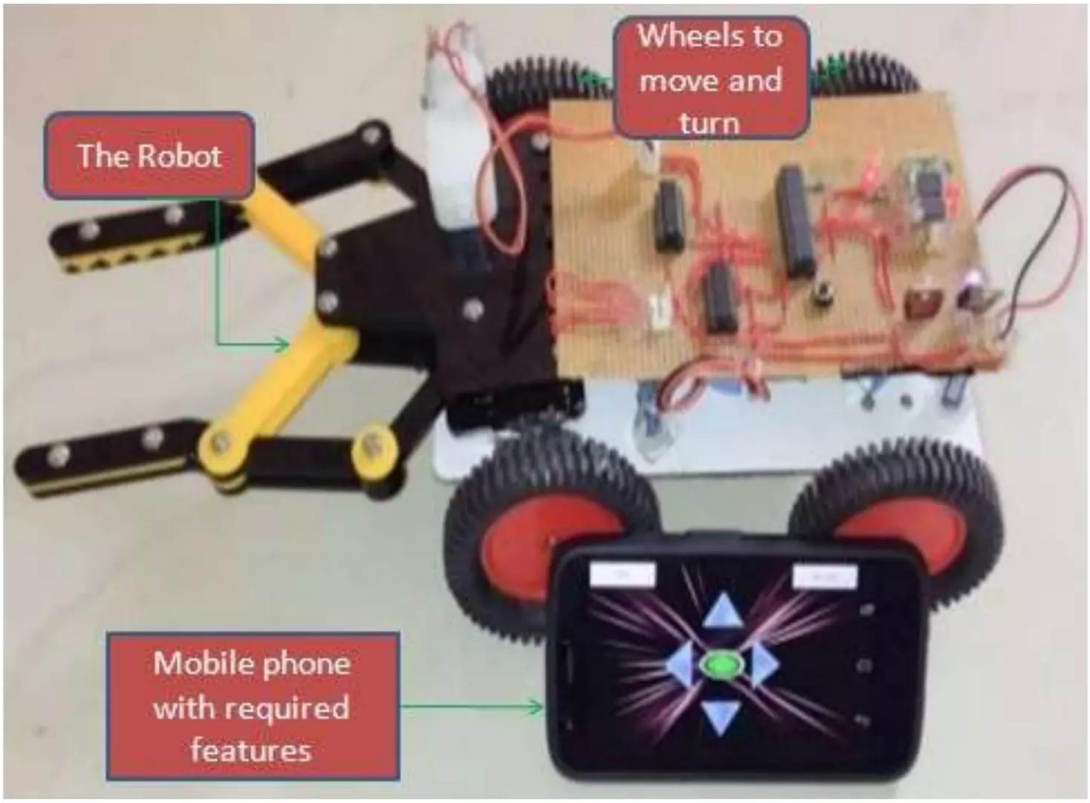
Bluetooth Controlled ROBOT

Aim of the project is to design a robot which can be controlled by Bluetooth of your mobile phone or PC.

An application program running in PC or android phone is used to send commands that are received by Bluetooth module attached with the robot.

Overview



Bluetooth Controlled Robot

Hardware components required and their purpose:

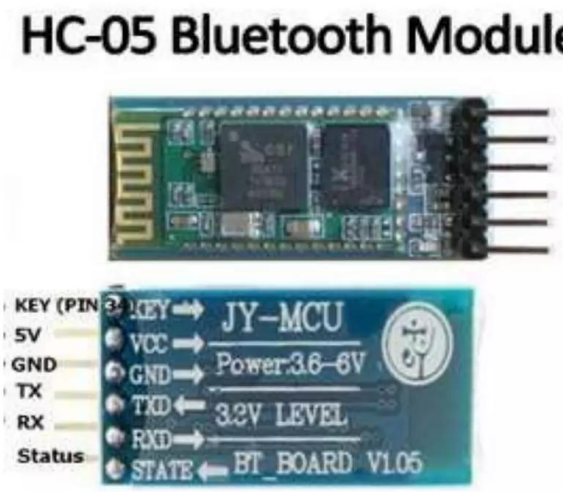
1. Arduino UNO board
2. PC or Mobile phone with customizable Bluetooth module
3. HC-05 Bluetooth module
4. DC motor
5. Motor driver IC (L293D)
6. Wheels
7. Power adopter

- Arduino UNO board: This is the brain of this robot in which the program is loaded to do the required functioning and is interfaced with Bluetooth module and the motor driver to make the system work as required.

- HC-05 Bluetooth Module: This module is capable of communicating with PC, mobile phone or any other Bluetooth enabled device. It is interfaced with the microcontroller over the

  serial UART port of micro-controller.

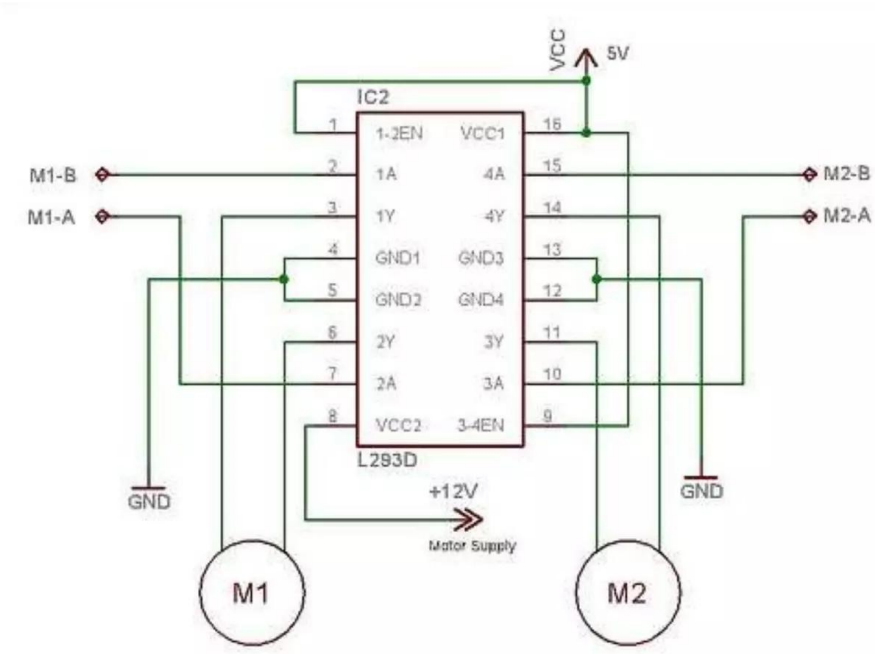
- PC or Mobile phone with customizable Bluetooth module: this works as the remote control for the robot. It has an application program running on it which enables us to send appropriate command over its Bluetooth module to control the robot.



Bluetooth Module

- DC Motor: This motor is controlled with DC voltages and can move in forward and backward direction according to the polarity of the voltage applied.

- Motor driver IC (L293D): Microcontrollers can't supply the current required by DC motor to run. So, to fulfill this requirement these motor driver ICs are used.

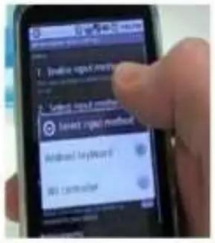


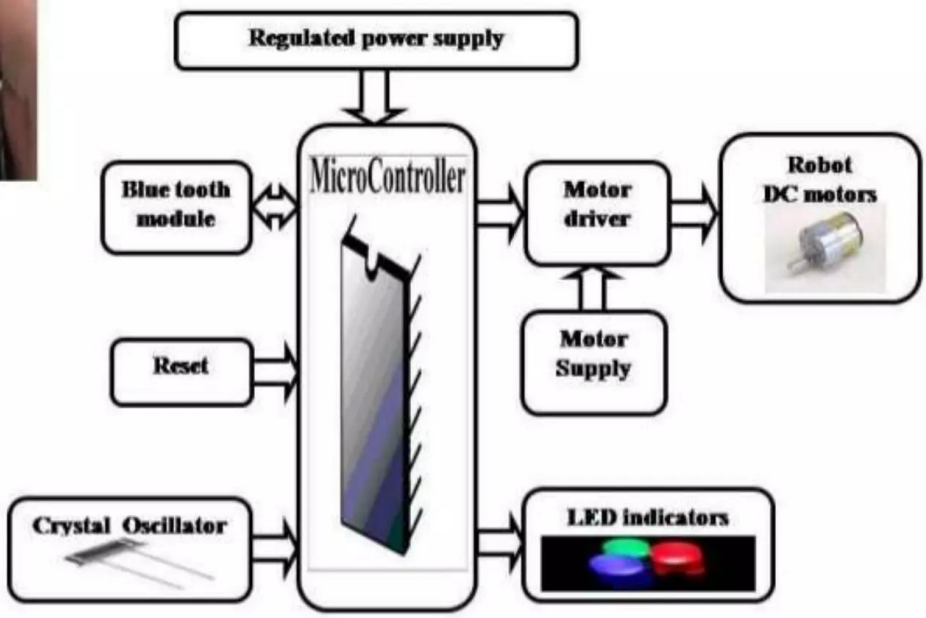
DC motors with Driver IC

- Power adopter: This is used to give appropriate dc power supply to microcontroller, driver IC sensors and the other passive components of the robot.

- Wheels: In it three wheels are employed, two at rear end and one at front end. Rear wheels are attached with the motors and also control the steering of robot. Front wheel is the loose steered wheel which moves in the direction of the pressure applied to it.

Block Diagram:





Bluetooth Controlled Robot

Description

For this firstly the application program is developed or a readymade program is made available to run on the PC or the mobile phone. This application program send the commands in form of ASCII characters which are received by HC-05 Bluetooth module and passed to microcontroller through UART port of microcontroller.

The microcontroller is programmed to take desired actions according to the command (ASCII character) received to move forward, reverse or to take a turn.

The microcontroller sends logic 1 or 0 at the specified pin to control motors of robot which are attached using motor driver IC (L293D).

Program:

```
/*
left motor attached to pin 5(+ve),6 and
right motor attached to pin 7(+ve),8 and
*/
void setup() {
 //Initialize serial
  Serial.begin(9600);


   pinMode(5, OUTPUT);
   pinMode(6, OUTPUT);
   pinMode(7, OUTPUT);
   pinMode(8, OUTPUT);
}
void loop()
{
char value;
value=Serial.read();


switch(value)
{
  case 'w':
  {
digitalWrite(6, LOW);
```

```
digitalWrite(8,  LOW);
digitalWrite(5,  HIGH);
digitalWrite(7,  HIGH);
  break;
  }
  case 'd':
  {
digitalWrite(5,  HIGH);
digitalWrite(7,  LOW);
digitalWrite(6,  LOW);
digitalWrite(8,  LOW);
  break;
    }
  case 'a':
  {
digitalWrite(5,  LOW);
digitalWrite(7,  HIGH);
digitalWrite(6,  LOW);
digitalWrite(8,  LOW);
  break;
  }

  case 'x':
  {
digitalWrite(5,  LOW);
digitalWrite(7,  LOW);
digitalWrite(6,  HIGH);
digitalWrite(8,  HIGH);
```

```
  break;
  }

 case ' ':
  {
digitalWrite(5,  LOW);
digitalWrite(7,  LOW);
digitalWrite(6,  LOW);
digitalWrite(8,  LOW);
  break;
  }
  }
}
```

Programming Digital I/O pins of Arduino UNO board:

- Each pin is controlled by three commands associated with it which are designated as:

pinMode()

digitalWrite()

digitalRead()

• pinMode()

This configures the specified pin to behave either as an input or an output.

Syntax

pinMode(pin, mode)

Parameters

pin: the number of the pin whose mode you wish to set

mode: INPUT, OUTPUT.

Returns

None

Example

```
int ledPin = 13;                // LED connected to digital pin 13

void setup()
{
  pinMode(ledPin, OUTPUT);      // sets the digital pin as output
}

void loop()
{
  digitalWrite(ledPin, HIGH);   // sets the LED on
  delay(1000);                  // waits for a second
  digitalWrite(ledPin, LOW);    // sets the LED off
```

```
  delay(1000);                // waits for a second
}
```

- **digitalWrite()**

Write a HIGH or a LOW value to a digital pin.

If the pin has been configured as an OUTPUT with pinMode(), its voltage will be set to the corresponding value: 5V (or 3.3V on 3.3V boards) for HIGH, 0V (ground) for LOW.

Syntax

digitalWrite(pin, value)

Parameters

pin: the pin number

value: HIGH or LOW

Returns

None

Example

Sets pin 13 to HIGH, makes a one-second-long delay, and sets the pin back to LOW.

```
int ledPin = 13;                // LED connected to digital pin 13
void setup()
{
  pinMode(ledPin, OUTPUT);      // sets the digital pin as output
}
void loop()
{
  digitalWrite(ledPin, HIGH);   // sets the LED on
  delay(1000);                  // waits for a second
```

```
 digitalWrite(ledPin, LOW);    // sets the LED off
 delay(1000);                  // waits for a second
}
```

- **digitalRead()**

Reads the value from a specified digital pin, either HIGH or LOW.

Syntax

digitalRead(pin)

Parameters

pin: the number of the digital pin you want to read (int)

Returns

HIGH or LOW

Example

```
 int ledPin = 13; // LED connected to digital pin 13
int inPin = 7;  // pushbutton connected to digital pin 7
int val = 0;    // variable to store the read value
void setup()
{
 pinMode(ledPin, OUTPUT);      // sets the digital pin 13 as output
 pinMode(inPin, INPUT);        // sets the digital pin 7 as input
}
void loop()
{
 val = digitalRead(inPin);    // read the input pin
 digitalWrite(ledPin, val);    // sets the LED to the button's value
}
```